

Zeineb SOUSSI

AI Engineer & Data Scientist

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in designing and deploying AI-driven applications that integrate data analysis, automation, and machine learning. Skilled in delivering measurable business outcomes through advanced AI techniques including LLMs, RAG, and multi-agent systems across the full product lifecycle.

EDUCATION

Engineering Degree, Computer Science — National School of Computer Science (ENSI), Tunis, Tunisia 2022–2025
Preparatory Engineering Studies — IPEIS, Sfax, Tunisia 2020–2022

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Zerda Digital Mar–Jul 2025

- Reduced content generation time by 95% through the development of a multi-agent AI marketing platform adopted in **20+ campaigns**.
- Cut manual validation time via a **RAG** architecture with semantic search (Tavily).
- Increased client engagement through HubSpot API personalization automated with **AWS Lambda & Bedrock**.
- Built a full-stack app (Django & LangChain) enabling **daily multi-format exports** across internal teams.

Technologies: AWS (Bedrock, Lambda), Django, LangChain, FastAPI, HubSpot API, Docker, NLP, RAG, GitLab

AI Engineer Intern — SlashUp Studio Feb–Mar 2025

- Accelerated support responses with a **RAG + LangChain** insurance chatbot for real-time Q&A.
- Improved query success rate by deploying **semantic search** over mixed policy datasets.

Technologies: Python, LangChain, RAG, OpenAI/LLM, Vector Databases, NLP, Semantic Search, GitLab

Software Engineer Intern — Innov-Alliance-Tech Jun–Aug 2024

- Scaled a Hadoop & Spark pipeline handling **large-scale IoT streams** for smart-city analytics.
- Enhanced reporting speed via dashboards for energy analytics and urban planning stakeholders.

Technologies: Hadoop, Apache Spark, Python, Big Data Analytics, GitHub

PROJECT EXPERIENCE

Project: Sentiment Analysis Image Classifier — ENSI, Tunis 2023

Achieved **92% accuracy** in real-time emotion recognition using **YOLOv5 & TensorFlow** on a Roboflow-based custom dataset.

Technologies: YOLOv5, TensorFlow, Roboflow

Project: Tunisian Sign Language Recognition — ENSI, Tunis 2024

Developed a computer-vision system for TSL translation with **88% accuracy** using **MediaPipe** hand detection & custom gesture recognition.

Technologies: MediaPipe, TensorFlow, YOLOv5

Project: Weather Prediction (Time Series) — ENSI, Tunis 2025

Designed forecasting models reaching **85% accuracy** using feature engineering and ensemble methods (**XGBoost**).

Technologies: Scikit-learn, XGBoost, Pandas

TECHNICAL SKILLS

AI/ML:	LLMs, LangChain, Multi-Agent Systems, RAG, Prompt Engineering, Deep Learning, CNN, RNN, LSTM, YOLO, NLP, TensorFlow, PyTorch, Computer Vision	Cloud:	AWS (Lambda, Bedrock)
Programming:	Python, SQL, TypeScript, Bash	Big Data:	Hadoop, Apache Spark, Kafka, Databricks
Frameworks:	Django, FastAPI, Flutter	Databases:	PostgreSQL, MongoDB, MySQL
		Methodologies:	Agile, Scrum, CRISP-DM

LANGUAGES

Arabic — Native French — Advanced English — Professional Working

LEADERSHIP & ACTIVITIES

Senior Member	Enactus ENSI	Mentored junior members in tech projects	2025
HR & Secretary	Enactus ENSI	Managed operations for 20+ members	2024
Active Member	ENSI Junior & ENSI Robotics	Contributed to robotics and engineering projects	2023